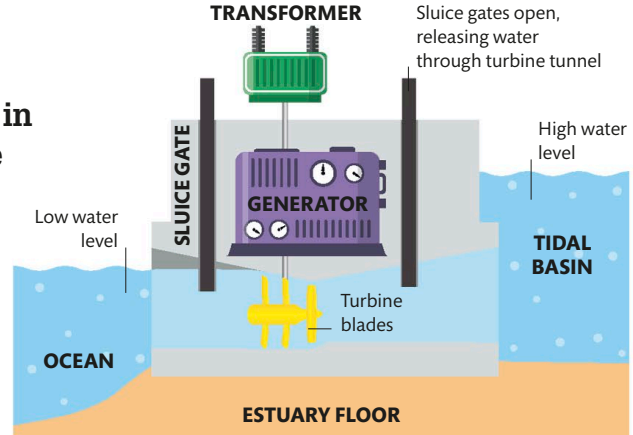
Water and geothermal power

The energy in moving water and the heat in Earth's crust can be harnessed to produce electricity. Both offer clean, sustainable sources of energy but tend to involve significant infrastructural investment.

Tidal power

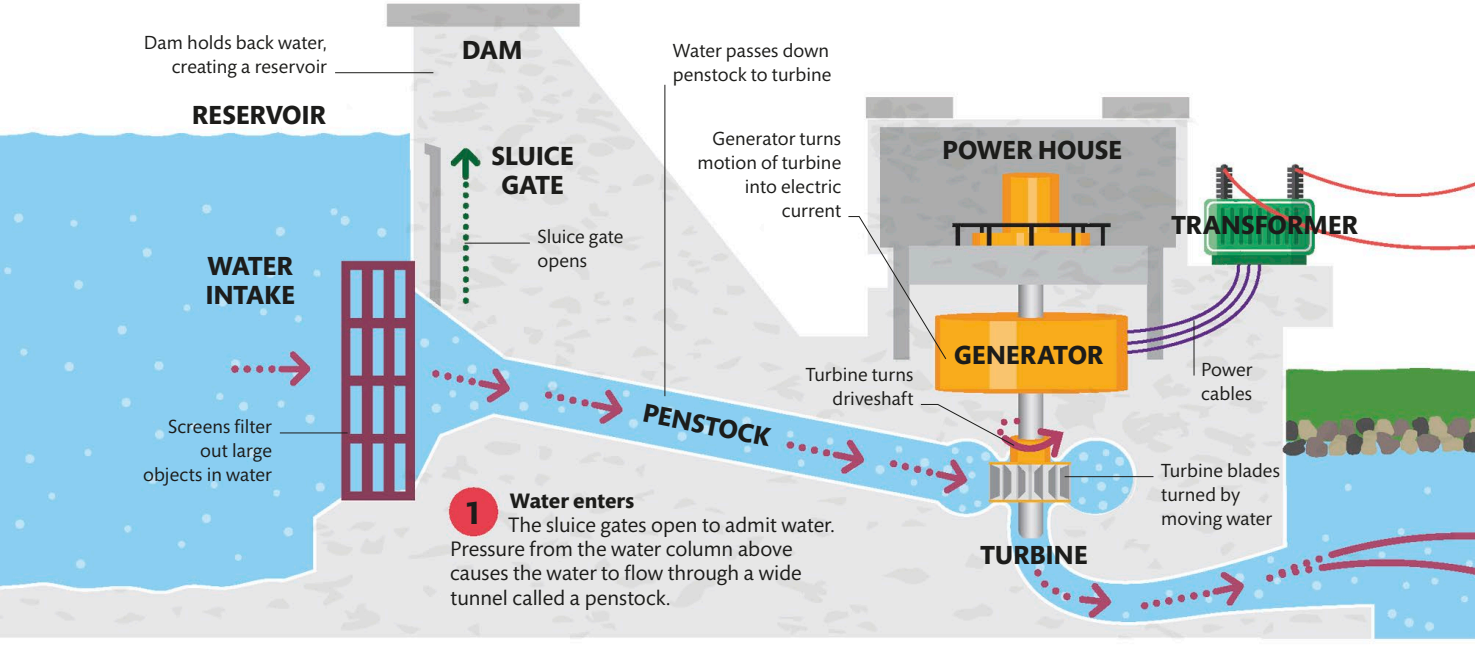
Tidal power captures the kinetic energy of the natural ebb and flow of ocean tides to turn turbines that power electricity generators. Some systems use freestanding turbines, similar to wind turbines, while tidal barrages involve multiple turbines in a large dam, usually constructed across a bay or estuary.



Tidal barrage
Water flowing into an estuary at high tide is held back behind a dam called a barrage. Sluice gates then control its flow through the turbine.

Hydroelectric power

Hydroelectric power (HEP) harnesses the power in falling or fast-flowing water to turn turbines that drive an electricity generator. Most commonly, water is collected at higher elevations behind a dam and then channeled past turbines.



1 Water enters
The sluice gates open to admit water. Pressure from the water column above causes the water to flow through a wide tunnel called a penstock.

2 Power generated
The water passes through a turbine at high speed, turning its blades with considerable force. The turbine powers a generator that produces an electric current.